

Software Architecture Document

DurandHealth



Version

1.0.0



Department

Engineering



Date

February 2025



Status

Active

Software Architecture and Design

1. Introduction

This Software Architecture Document (SAD) provides a detailed representation of the system's architecture, developed based on the functional and non-functional requirements provided by the user. It illustrates how these requirements shape the architectural design, ensuring the system meets its intended purpose effectively. The document incorporates various architecture diagrams that visually represent the structure, components, and interactions within the system. These diagrams are created iteratively during the RUP Elaboration phase, reflecting key decisions and ensuring alignment with user expectations and system constraints. Each section of the SAD builds upon the requirements, progressively detailing the architectural vision.

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1.1 Purpose

The purpose of this Software Architecture Document (SAD) is to provide a clear and comprehensive representation of the system's architecture, serving as a bridge between user requirements and technical implementation. It translates the functional and non-functional requirements provided by the user into a structured architectural framework, guiding the development process and ensuring alignment with stakeholder expectations.

1.2 Scope

The scope of this Software Architecture Document (SAD) is to define and communicate the system architecture based on the functional and non-functional requirements gathered from the user. It serves as a blueprint for stakeholders, providing clarity on how the system's components, modules, and interactions are designed to meet the defined requirements.

1.3 Definitions, Acronyms and Abbreviations

RUP: Rational Unified Process UML: Unified Modeling Language SAD: Software Architecture Document

2. Architectural Representations

Conceptual, Logical and Physical Views

The SAD employs a stakeholder-focused, multiple view approach to architecture documentation, as required by ANSI/IEEE 1471-2000, the recommended best practice for documenting the architecture of software-intensive systems [IEEE 1471]. As described in Section ??, a software architecture comprises more than one software structure, each of which provides an engineering handle on different system qualities. A view is the specification of one or more of these structures, and documenting a software architecture, then, is a matter of documenting the relevant views and then documenting information that applies to more than one view [Clements 2002]. ANSI/IEEE 1471-2000 provides guidance for choosing the best set of views to document, by bringing stakeholder interests to bear. It prescribes defining a set of viewpoints to satisfy the stakeholder community. A viewpoint identifies the set of concerns to be addressed, and identifies the modeling techniques, evaluation techniques, consistency checking techniques, etc., used by any conforming view. A view, then, is a viewpoint applied to a system. It is a representation of a set of software elements, their properties, and the relationships among them that conform to a defining viewpoint. Together, the chosen set of views show the entire architecture and all of its relevant properties. A SAD contains the viewpoints, relevant views, and information that applies to more than one view to give a holistic description of the system.

3. Functional Requirements

1. Implement user and identity management with role-based access control, SSO support, secure access via Identity-Aware Proxy, and HIPAA-compliant consent workflows.
2. The Analytics & Insights Module shall provide dashboards for risk modeling and user engagement analytics.
3. The system shall dynamically onboard patients and build personalized care plans based on their health goals and history, utilizing Firestore for data management.
4. Implement a data storage and compliance layer using Firestore, Cloud SQL, and Cloud Storage to ensure Assured Workloads and HIPAA compliance.
5. The Corporate Portal shall provide dashboards for HR teams to track employee health engagement and manage incentive programs.
6. The system shall manage digital consents with features for revocation and audit tracking.
7. The system shall provide a dynamic AI-supported health risk questionnaire and a multimodal service booking platform for users to assess health risks and schedule services.
8. The system shall provide individual users with personalized health assessments, multimodal service booking, AI-driven follow-up prompts, and supplement refill alerts.
9. The system shall provide AI-driven care plan and recommendations based on user behavior, history, and goals using OpenAI APIs.
10. The system shall enable corporate users to track benefit usage, analyze wellness trends, personalize health campaigns, and manage consent governance through comprehensive reporting and audit logs.
11. The system shall provide a virtual assistant for health literacy, reminders, and behavior change, focusing on non-PHI interactions.

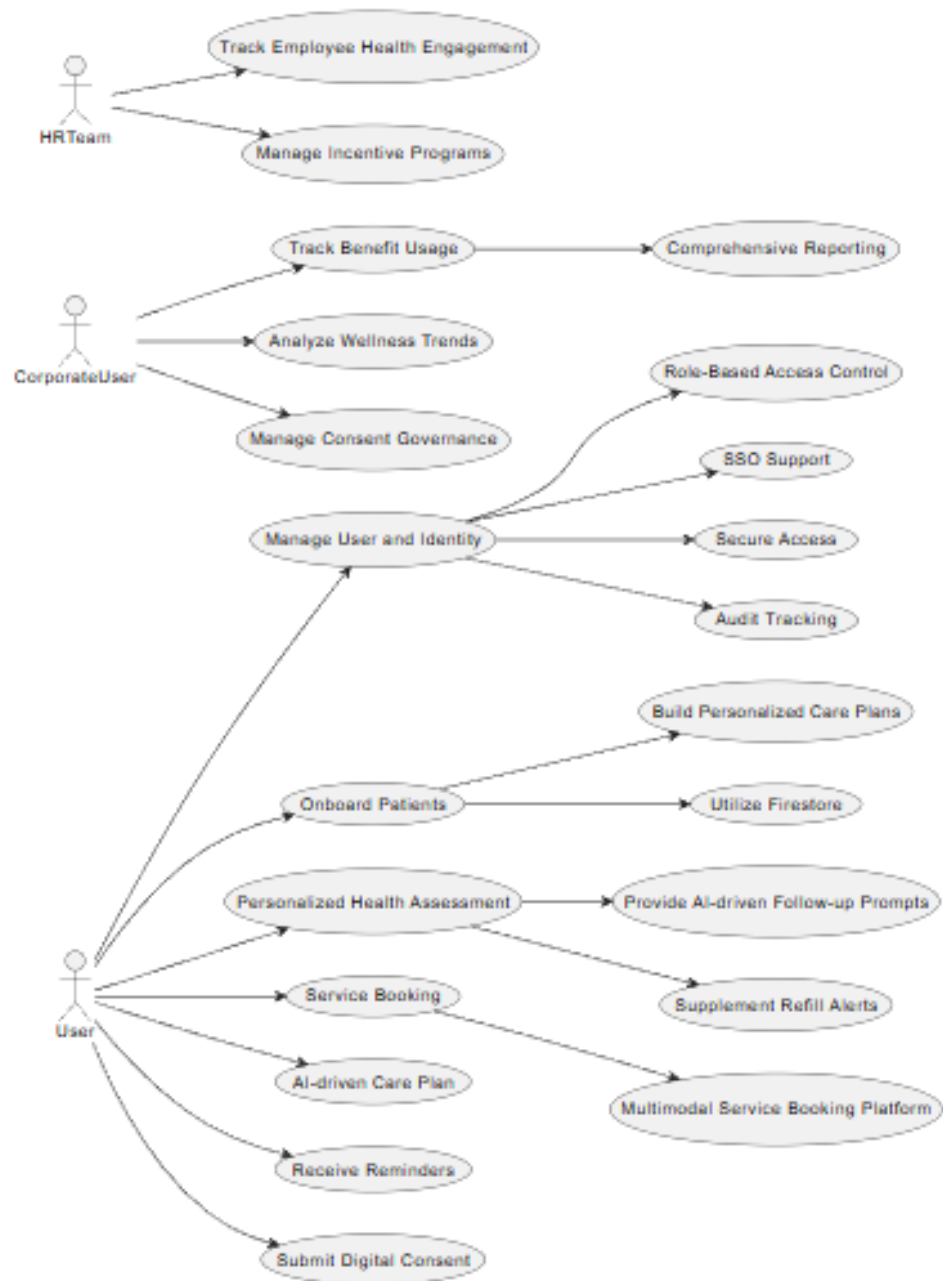
4. Non-Functional Requirements

1. The user and identity management system must ensure role-based access control is enforced within 100 milliseconds for each access request.
2. The system must support Single Sign-On (SSO) with an uptime of 99.9% to ensure continuous access.
3. Secure access via Identity-Aware Proxy must be implemented with end-to-end encryption for all data in transit.
4. The system must comply with HIPAA regulations for data handling, ensuring all consent workflows are auditable and traceable.
5. The Analytics & Insights Module must generate dashboards with data refresh rates not exceeding 5 minutes to provide timely insights.
6. The system must be capable of onboarding up to 1,000 patients concurrently without degradation in performance.
7. The data storage layer must ensure a minimum availability of 99.95% for Firestore, Cloud SQL, and Cloud Storage.
8. The Corporate Portal must load user dashboards within 3 seconds for optimal user experience.
9. The system must provide audit tracking for digital consents with a retention period of at least 6 years to meet compliance standards.
10. The AI-supported health risk questionnaire must process user responses within 2 seconds to maintain user engagement.
11. The multimodal service booking platform must achieve a success rate of at least 98% for scheduled appointments.
12. The system must provide personalized health assessments with an accuracy of at least 90% based on user data.
13. The virtual assistant must respond to user inquiries within 1 second to ensure a seamless interaction experience.
14. The system must handle up to 10,000 simultaneous users without any performance degradation.
15. The system must ensure that all AI-driven recommendations are generated with a confidence level of at least 85%.

16. The system must provide comprehensive reporting and audit logs with a generation time not exceeding 5 seconds for corporate users.

5. Use Case Diagram

A use-case model is a visual representation that outlines all the use cases of a system and their interactions with users, called actors. It typically includes use case diagrams that depict the actors and the interactions they have with the system's functionalities. The model uses symbols like ovals for use cases and stick figures for actors. It also includes a system boundary to define the scope and various lines or arrows to show relationships between actors and use cases. The purpose of the use-case model is to clearly represent how users interact with the system and to help guide the design, development, and testing processes by providing a shared understanding of system requirements among stakeholders like business analysts, developers, and clients.



5.1. Use Case SpecificationsReceive RemindersDescription :-

- This use case outlines the process by which a user receives reminders for scheduled events or tasks through their preferred notification method.

Actors :-

- User
- Reminder System

Preconditions :-

- The user has registered for the reminder service.
- The user has set up reminders in the system.

Steps :-

1. Trigger Reminder: The system checks for any reminders that are due based on the user's set schedule.
2. Prepare Notification: The system formats the reminder message according to user preferences (e.g., email, SMS, app notification).
3. Send Notification: The system sends the reminder to the user through the chosen notification channel.
4. User Acknowledges: The user receives the reminder and may choose to acknowledge or act on it.
5. Log Activity: The system logs the reminder delivery and any user interaction for future reference.

Exceptions :-

1. Notification Failure: If the reminder fails to send (e.g., due to network issues), the system logs the error and may attempt to resend.
2. User Unsubscribed: If the user has unsubscribed from reminders, the system does not send notifications.

Post-conditions :-

- The user is informed of the scheduled reminder.
- The system logs the reminder delivery and user interactions for auditing and improvements.

5.2. Use Case SpecificationsService BookingDescription :-

- This use case describes the process by which a customer books a service through an online platform, selecting options, providing necessary information, and confirming the booking.

Actors :-

- Customer
- Service Booking System

Preconditions :-

- The customer has an active account and is logged into the system.
- The service booking system is available and functioning properly.

Steps :-

1. Initiate Booking: The customer navigates to the service booking page.
2. Select Service: The customer selects the desired service from the available options.
3. Choose Date and Time: The customer selects a preferred date and time for the service.
4. Provide Information: The customer enters any required personal information and service details.
5. Review Booking: The customer reviews the booking details for accuracy.
6. Confirm Booking: The customer confirms the booking, and the system processes the request.
7. Receive Confirmation: The system sends a confirmation message or email to the customer.

Exceptions :-

1. Service Unavailable: If the selected service is not available at the chosen time, the system prompts the customer to select a different time.
2. Invalid Input: If the customer enters invalid information, the system displays an error message and requests correction.
3. Payment Issues: If payment is required and fails, the system notifies the customer to retry payment.

Post-conditions :-

- The service is successfully booked for the customer.
- The system updates the schedule to reflect the new booking.

5.3. Use Case Specifications Personalized Health Assessment Description :-

- This use case describes the process by which a user completes a personalized health assessment through an online system, receiving tailored health recommendations based on their inputs.

Actors :-

- User
- Health Assessment System

Preconditions :-

- The user has created an account and is logged into the system.
- The system is available and functioning properly.

Steps :-

1. Initiate Assessment: The user navigates to the health assessment section of the system.
2. Provide Personal Information: The user enters their personal details (age, gender, weight, height).
3. Answer Health Questions: The user responds to a series of health-related questions (medical history, lifestyle habits).
4. Submit Assessment: The user submits their completed assessment for evaluation.
5. Generate Results: The system analyzes the input data and generates personalized health recommendations.
6. Display Recommendations: The system presents the user with their health assessment results and recommendations.
7. Option to Save or Share: The user can save the results or share them with a healthcare provider.

Exceptions :-

1. Incomplete Submission: If the user fails to fill out required fields, the system prompts them to complete the assessment.
2. Technical Error: If there is a system malfunction, the user is notified and asked to try again later.

Post-conditions :-

- The user receives personalized health recommendations.
- The assessment results are saved in the user's account for future reference.

5.4. Use Case Specifications AI-driven Care Plan Description :-

- This use case describes how an AI system generates a personalized care plan for a patient based on their health data, preferences, and clinical guidelines.

Actors :-

- Patient
- Healthcare Provider
- AI System

Preconditions :-

- The patient has provided necessary health information and consent for AI analysis.
- The AI system is operational and has access to the patient's medical history.

Steps :-

1. Patient Data Input: The healthcare provider inputs the patient's health data into the AI system.
2. Data Analysis: The AI system analyzes the patient's data, including medical history and current health status.
3. Care Plan Generation: The AI generates a personalized care plan that includes treatment recommendations, lifestyle changes, and follow-up schedules.
4. Review Care Plan: The healthcare provider reviews the AI-generated care plan for accuracy and relevance.
5. Patient Consultation: The healthcare provider discusses the care plan with the patient, addressing any concerns.
6. Finalize Plan: The provider finalizes the care plan and shares it with the patient, ensuring understanding and agreement.

Exceptions :-

1. Incomplete Data: If the patient's data is incomplete, the system prompts the provider to enter missing information.
2. Plan Disagreement: If the healthcare provider disagrees with the AI's recommendations, they may modify the plan or consult additional resources.

Post-conditions :-

- A personalized care plan is created and agreed upon by the patient and healthcare provider.
- The patient receives the care plan for implementation and follow-up.

5.5. Use Case Specifications Onboard Patients Description :-

- This use case details the process of onboarding new patients into a healthcare system, ensuring that their personal information is collected, stored, and verified.

Actors :-

- Patient
- Healthcare Provider
- Patient Registration System

Preconditions :-

- The patient has provided necessary personal and medical information.
- The healthcare provider has access to the patient registration system.

Steps :-

1. Initiate Onboarding: The patient accesses the patient registration system.
2. Enter Personal Information: The patient inputs their personal details, such as name, date of birth, and contact information.
3. Provide Medical History: The patient enters relevant medical history and current health concerns.
4. Verify Information: The system reviews the provided information for completeness and accuracy.
5. Create Patient Profile: The system creates a new patient profile based on the entered information.
6. Confirm Registration: The patient receives confirmation of successful onboarding, along with next steps or appointments.

Exceptions :-

1. Incomplete Information: If required fields are missing, the system prompts the patient to complete all necessary sections.
2. Invalid Data: If the data format is incorrect (e.g., invalid date), the system alerts the patient to correct the information.

Post-conditions :-

- The patient is successfully onboarded into the healthcare system.
- The patient profile is stored securely in the system for future reference.

5.6. Use Case SpecificationsManage User and IdentityDescription :-

- This use case outlines how users can manage their identities, including creating, updating, and deleting user accounts, as well as managing authentication methods.

Actors :-

- User
- Identity Management System
- Administrator

Preconditions :-

- The user has an account or is attempting to create one.
- The identity management system is operational and accessible.

Steps :-

1. User Login: The user accesses the identity management system and logs in with credentials.
2. View Account: The user navigates to the user management section to view account details.
3. Update Information: The user selects the option to update personal information (e.g., email, password).
4. Save Changes: The user submits the changes, and the system validates and updates the account.
5. Account Deletion: The user can choose to delete their account, confirming the action.
6. Confirmation: The system confirms the successful update or deletion of the account.

Exceptions :-

1. Invalid Credentials: If the user provides incorrect login details, the system prompts for re-entry.
2. Update Failure: If the system encounters an error while updating, it notifies the user with an error message.
3. Account Deletion Denied: If the user cannot delete their account due to policy restrictions, the system informs them.

Post-conditions :-

- The user's account information is updated or deleted as requested.
- The system logs the changes made to the user's identity for auditing purposes.

5.7. Use Case Specifications Submit Digital Consent Description :-

- This use case describes the process by which a user submits their digital consent for a specific purpose, ensuring compliance with legal and regulatory requirements.

Actors :-

- User
- Digital Consent Management System

Preconditions :-

- The user has an active account and is logged into the system.
- The digital consent management system is operational and accessible.

Steps :-

1. Navigate to Consent Section: The user accesses the digital consent section of the application.
2. Review Consent Details: The system displays the consent information, including terms and conditions.
3. Accept Terms: The user reads and accepts the terms by checking a consent box.
4. Submit Consent: The user clicks the 'Submit' button to send their consent.
5. Confirmation: The system processes the consent and displays a confirmation message.
6. Record Consent: The system saves the consent information in the user's profile.

Exceptions :-

1. Incomplete Information: If the user fails to accept the terms, the system prompts the user to complete the action.
2. Submission Error: If there's a technical issue, the system alerts the user and prompts them to retry.

Post-conditions :-

- The user's digital consent is recorded in the system.
- The system confirms the user's consent submission.

5.8. Use Case SpecificationsTrack Employee Health EngagementDescription :-

- This use case outlines how an employee can track their health engagement activities, such as fitness challenges, wellness programs, and health assessments, within the health engagement system.

Actors :-

- Employee
- Health Engagement System
- HR Administrator

Preconditions :-

- The employee is registered in the health engagement system.
- The health engagement system is operational and accessible.

Steps :-

1. Log In: The employee logs into the health engagement system using their credentials.
2. Access Dashboard: The system displays the employee's health engagement dashboard.
3. View Activities: The employee selects the option to view their current health engagement activities and programs.
4. Record Activity: The employee records new health engagement activities, such as workouts or wellness program participation.
5. Review Progress: The employee reviews their progress and engagement metrics provided by the system.
6. Set Goals: The employee sets new health engagement goals based on their progress.
7. Log Out: The employee logs out of the system.

Exceptions :-

1. Invalid Login: If the employee enters incorrect credentials, the system prompts them to retry.
2. System Down: If the health engagement system is not available, the employee receives an error message.

Post-conditions :-

- The employee's recorded health activities are saved in the system.
- The employee's health engagement metrics are updated.

5.9. Use Case Specifications

Analyze Wellness Trends

Description :-

- This use case outlines the process by which a health analyst reviews and analyzes data related to wellness trends among users to derive insights and recommendations.

Actors :-

- Health Analyst
- Wellness Data System
- End Users (participants in wellness programs)

Preconditions :-

- The wellness data system is operational and has access to relevant health and wellness data.
- Users have consented to share their wellness data.

Steps :-

1. Access System: The health analyst logs into the wellness data system.
2. Select Data Parameters: The analyst selects relevant wellness data parameters (e.g., physical activity, nutrition, mental health).
3. Generate Report: The system generates a report based on the selected parameters.
4. Analyze Data: The analyst reviews the report for trends, patterns, and anomalies.
5. Document Findings: The analyst documents insights and potential recommendations based on the analysis.
6. Share Insights: The analyst shares findings with relevant stakeholders through the system.

Exceptions :-

1. Data Unavailable: If the requested data is not available, the system alerts the analyst and suggests alternative data sources.
2. Access Denied: If the analyst lacks permissions to access certain data, the system notifies them of insufficient access rights.

Post-conditions :-

- The wellness trends report is generated and available for review.

- Insights and recommendations are documented and shared with stakeholders.

5.10. Use Case SpecificationsManage Consent GovernanceDescription :-

- This use case describes the process by which data stewards manage and govern consent provided by data subjects for the use of their personal data, ensuring compliance with privacy regulations.

Actors :-

- Data Steward
- Consent Management System
- Data Subject

Preconditions :-

- The data subject has provided personal data to the organization.
- The consent management system is operational and accessible.

Steps :-

1. Access Consent Management System: The data steward logs into the consent management system.
2. Review Existing Consents: The system displays a list of existing consents provided by data subjects.
3. Update Consent: The data steward selects a consent record to update and modifies the consent terms as required.
4. Validate Changes: The system checks the validity of the updated consent terms against regulatory requirements.
5. Notify Data Subject: The system sends a notification to the data subject informing them of the changes made to their consent.
6. Record Changes: The updated consent is saved in the system, and an audit log is created.

Exceptions :-

1. Invalid Consent Terms: If the updated terms are invalid, the system prompts the data steward to correct them.
2. Notification Failure: If the notification to the data subject fails, the system alerts the data steward.

Post-conditions :-

- The consent record is updated successfully in the system.
- The data subject is informed of the changes to their consent.

5.11. Use Case Specifications Role-Based Access Control Description :-

- This use case describes the process of managing user access rights and permissions based on their assigned roles within an organization.

Actors :-

- User
- Access Control System
- Administrator

Preconditions :-

- The user has a valid account in the system.
- The access control system is operational and configured.

Steps :-

1. User Login: The user logs into the system with their credentials.
2. Role Assignment: The administrator assigns roles to the user based on their job function.
3. Access Configuration: The system configures access rights for the user based on their role.
4. Access Request: The user requests access to a specific resource or application.
5. Access Verification: The access control system checks the user's role and permissions against the requested resource.
6. Access Grant/Denial: The system grants or denies access based on the verification outcome.
7. Log Activity: The system logs the access request and its outcome for auditing purposes.

Exceptions :-

1. Invalid Credentials: If the user provides incorrect login information, the system prompts for re-entry.
2. Role Not Assigned: If the user does not have a role assigned, access requests are denied with a notification.
3. System Error: If the access control system encounters an error, the user is notified and the request is logged.

Post-conditions :-

- The user has the appropriate access rights based on their assigned role.
- Access requests are logged for future auditing and compliance.

5.12. Use Case SpecificationsTrack Benefit UsageDescription :-

- This use case outlines the process by which a user can track their benefit usage within a system, allowing them to view, manage, and analyze their benefits over time.

Actors :-

- User
- Benefit Tracking System

Preconditions :-

- The user has an active account and is logged into the system.
- The benefit tracking system is operational and accessible.

Steps :-

1. Login: The user logs into the benefit tracking system using their credentials.
2. Access Benefit Dashboard: The user navigates to the benefit usage dashboard.
3. View Benefit Overview: The system displays an overview of available benefits and their usage statistics.
4. Select Benefit for Details: The user selects a specific benefit to view detailed usage information.
5. Analyze Usage: The system presents detailed usage metrics, including dates, amounts, and remaining balances.
6. Set Notifications: The user opts to set notifications for usage thresholds or expirations.
7. Logout: The user logs out of the system after reviewing their benefits.

Exceptions :-

1. Unauthorized Access: If the user attempts to access the system without valid credentials, an error message is displayed.
2. Data Unavailable: If benefit usage data is not available, the system notifies the user with an appropriate message.

Post-conditions :-

- The user has successfully tracked their benefit usage.

- Notification preferences are updated if set by the user.

5.13. Use Case Specifications Secure Access Description :-

- This use case describes the process by which a user requests access to a secure system, authenticates their identity, and gains access to the resources.

Actors :-

- User
- Secure Access System

Preconditions :-

- The user has valid credentials (username and password).
- The secure access system is operational and accessible.

Steps :-

1. Access Login Page: The user navigates to the secure access system's login page.
2. Enter Credentials: The user inputs their username and password.
3. Submit Credentials: The user submits their credentials for verification.
4. Authenticate User: The system verifies the user's credentials against the database.
5. Grant Access: If the credentials are valid, the system grants access and redirects the user to the secure resources.
6. Access Resources: The user interacts with the secure resources as permitted.

Exceptions :-

1. Invalid Credentials: If the username or password is incorrect, the system displays an error message and prompts for re-entry.
2. Account Locked: If the account is locked due to multiple failed attempts, the system informs the user and suggests recovery options.
3. System Error: If there is a technical issue, the system provides an error message and advises the user to try again later.

Post-conditions :-

- The user is successfully authenticated and granted access to the secure system.

- The system logs the access attempt for security auditing.

5.14. Use Case SpecificationsManage Incentive ProgramsDescription :-

- This use case outlines the process for creating, updating, and managing incentive programs designed to motivate and reward participants within an organization.

Actors :-

- Program Manager
- Incentive Program System
- Participants

Preconditions :-

- The Program Manager has the necessary permissions to manage incentive programs.
- The Incentive Program System is operational and accessible.

Steps :-

1. Log In: The Program Manager logs into the Incentive Program System.
2. Access Program Management: The Program Manager navigates to the incentive program management section.
3. Create New Program: The Program Manager selects the option to create a new incentive program and fills in the required details (name, description, criteria).
4. Define Incentives: The Program Manager specifies the incentives associated with the program.
5. Set Eligibility Criteria: The Program Manager establishes eligibility requirements for participants.
6. Save Program: The Program Manager saves the new program, and the system confirms the creation.
7. Edit Existing Program: The Program Manager can select an existing program to modify details or incentives.
8. Review Program Performance: The Program Manager reviews the performance metrics of active programs.
9. Deactivate Program: The Program Manager can deactivate any incentive program that is no longer active.

Exceptions :-

1. Invalid Input: If any input data is invalid during program creation, the system prompts the Program Manager to correct it.
2. Unauthorized Access: If the Program Manager does not have the necessary permissions, the system denies access to program management functionalities.
3. System Error: If the system encounters a technical issue, it displays an error message and suggests trying again later.

Post-conditions :-

- The incentive program is created or updated in the system.
- Participants are notified of new or updated programs as applicable.

5.15. Use Case SpecificationsSSO SupportDescription :-

- This use case describes the process by which a user accesses an application using Single Sign-On (SSO) authentication through an identity provider.

Actors :-

- User
- Identity Provider
- Application System

Preconditions :-

- The user has a valid account with the identity provider.
- The application system supports SSO integration.
- The user is attempting to access a resource that requires authentication.

Steps :-

1. Access Application: The user navigates to the application requiring authentication.
2. Redirect to Identity Provider: The application redirects the user to the identity provider's login page.
3. User Authentication: The user enters their credentials on the identity provider's login page.
4. Validate Credentials: The identity provider validates the user's credentials.
5. Generate Token: Upon successful validation, the identity provider generates an authentication token.
6. Redirect Back: The identity provider redirects the user back to the application with the token.
7. Token Validation: The application validates the received token with the identity provider.
8. Access Granted: If the token is valid, the user is granted access to the application.

Exceptions :-

1. Invalid Credentials: If the user enters incorrect credentials, the identity provider prompts them to retry.

2. Token Expired: If the token has expired, the application redirects the user back to the identity provider for re-authentication.

3. Service Unavailable: If the identity provider is down, the application displays an error message.

Post-conditions :-

- The user is authenticated and granted access to the application.
- The application logs the authentication event for auditing purposes.

5.16. Use Case Specifications

Audit Tracking

Description :-

- This use case outlines the process of tracking, recording, and reviewing user activities within a system to ensure compliance and accountability.

Actors :-

- System Administrator
- Audit Tracking System
- User

Preconditions :-

- The system administrator has the necessary permissions to access the audit tracking features.
- The audit tracking system is operational and has access to the relevant data.

Steps :-

1. Initiate Audit Tracking: The system administrator accesses the audit tracking interface.
2. Configure Tracking Parameters: The administrator sets parameters such as timeframes, user roles, and types of actions to be tracked.
3. Start Tracking: The administrator activates the tracking feature, enabling the system to record user activities.
4. Monitor Activities: The system automatically logs user actions, generating records that include timestamps, user IDs, and action types.
5. Generate Reports: The administrator requests a report of tracked activities based on predefined filters.
6. Review Audit Logs: The administrator reviews the generated report for any anomalies or compliance issues.
7. Take Action: If any issues are found, the administrator may take necessary corrective actions or escalate them.

Exceptions :-

1. Access Denied: If the administrator lacks sufficient permissions, access to audit tracking features is denied.

2. Data Retrieval Failure: If the system encounters issues retrieving audit data, an error message is displayed.

3. Report Generation Error: If there are problems generating the report, the system notifies the administrator.

Post-conditions :-

- User activities are recorded and available for review.
- Compliance reports are generated and stored for future reference.

5.17. Use Case SpecificationsSupplement Refill AlertsDescription :-

- This use case describes how the pharmacy system alerts patients when their supplements are due for refill based on their prescribed schedule.

Actors :-

- Patient
- Pharmacy System
- Healthcare Provider

Preconditions :-

- The patient has a registered account with the pharmacy system.
- The patient's medication history is accessible in the pharmacy system.

Steps :-

1. Monitor Medication Schedule: The pharmacy system regularly checks the medication schedule for each patient.
2. Identify Due Supplements: The system identifies which supplements are approaching their refill date.
3. Generate Alert: The system generates a refill alert for the identified supplements.
4. Notify Patient: The system sends a notification to the patient via email or app notification about the refill.
5. Patient Acknowledgment: The patient acknowledges the alert and can choose to order a refill directly from the notification.
6. Process Refill Order: If the patient opts to refill, the system processes the order and updates the inventory.

Exceptions :-

1. No Supplements Due: If no supplements are due for refill, no alert is generated.
2. Notification Failure: If the notification fails to send, the system logs the error for further investigation.

Post-conditions :-

- The patient is informed of upcoming supplement refills.
- The pharmacy system updates the medication schedule as required.

5.18. Use Case Specifications Provide AI-driven Follow-up Prompts Description :-

- This use case outlines how the AI system generates personalized follow-up prompts for the user based on prior interactions to enhance engagement and gather additional information.

Actors :-

- User
- AI System

Preconditions :-

- The user has access to the AI-driven system.
- The user has previously engaged in a conversation or activity that requires follow-up prompts.

Steps :-

1. User Interaction: The user completes a task or engages in a conversation with the AI system.
2. Context Analysis: The AI system analyzes the context and content of the user's recent interactions.
3. Prompt Generation: The AI system generates relevant follow-up prompts tailored to the user's needs.
4. Display Prompts: The AI system presents the follow-up prompts to the user for review.
5. User Response: The user selects a prompt to respond to or engages with the system based on the suggestions.
6. Process Response: The AI system processes the user's response and continues the interaction accordingly.

Exceptions :-

1. No Context Available: If there is insufficient context, the system notifies the user and may ask for clarification.
2. Prompt Generation Failure: If the AI fails to generate prompts, the system informs the user and may suggest generic follow-ups.

Post-conditions :-

- The user receives personalized follow-up prompts.

- The interaction history is updated to include the user's responses and selections.

5.19. Use Case SpecificationsBuild Personalized Care PlansDescription :-

- This use case outlines the process of creating tailored care plans for patients based on their individual health needs, preferences, and treatment goals.

Actors :-

- Patient
- Healthcare Provider
- Care Plan Management System

Preconditions :-

- The patient has been assessed and diagnosed by a healthcare provider.
- The healthcare provider is authenticated and authorized to access the care plan system.

Steps :-

1. Initiate Care Plan Creation: The healthcare provider accesses the care plan management system.
2. Patient Information Review: The provider reviews the patient's medical history and current health status.
3. Set Goals: The provider collaborates with the patient to identify specific health goals and objectives.
4. Select Interventions: The provider selects appropriate interventions and resources based on the patient's needs.
5. Document Care Plan: The provider inputs the care plan details into the system, including timelines and responsibilities.
6. Review and Approve: The provider reviews the care plan with the patient for approval.
7. Finalize Care Plan: Once approved, the system saves the care plan and notifies relevant healthcare team members.

Exceptions :-

1. Incomplete Patient Data: If patient data is incomplete, the system prompts the provider to gather necessary information.
2. Patient Declines Interventions: If the patient declines certain interventions, the provider must revise the care plan accordingly.

Post-conditions :-

- A personalized care plan is created and saved in the system.
- The healthcare team members are informed of the new care plan for the patient.

5.20. Use Case Specifications Utilize FirestoreDescription :-

- This use case outlines the process of interacting with a Firestore database, including creating, reading, updating, and deleting documents within collections.

Actors :-

- Developer
- Firestore Database

Preconditions :-

- The developer has access to a Firestore project and the necessary API credentials.
- The Firestore database is set up and accessible.

Steps :-

1. Initialize Firestore: The developer initializes the Firestore client using the provided API credentials.
2. Create Document: The developer creates a new document in a specified collection with relevant data.
3. Read Document: The developer retrieves a document from a collection using its unique identifier.
4. Update Document: The developer modifies an existing document's fields with new data.
5. Delete Document: The developer removes a document from the collection using its unique identifier.
6. Query Data: The developer performs a query to retrieve documents that match certain criteria.
7. Handle Errors: The developer checks for errors during each operation and handles them appropriately.

Exceptions :-

1. Document Not Found: If the document to read, update, or delete does not exist, an error is returned.
2. Permission Denied: If the developer lacks the necessary permissions, an error is returned.
3. Network Issues: If there are connectivity issues, operations may fail or time out.

Post-conditions :-

- The specified document is created, read, updated, or deleted in the Firestore database.
- The database reflects the latest changes made by the developer.

5.21. Use Case Specifications Comprehensive Reporting Description :-

- This use case outlines the process by which a user generates comprehensive reports using the reporting system, analyzing data across various parameters and formats.

Actors :-

- Business Analyst
- Reporting System
- Data Analyst

Preconditions :-

- The reporting system is operational and accessible.
- The user has appropriate permissions to access reporting features.
- Relevant data is available in the system for reporting.

Steps :-

1. Log In: The user logs into the reporting system with valid credentials.
2. Select Reporting Module: The user navigates to the reporting module of the system.
3. Choose Report Type: The user selects the type of report they wish to generate (e.g., sales, performance).
4. Define Parameters: The user specifies parameters for the report, such as date range, departments, or categories.
5. Generate Report: The user requests the system to generate the report.
6. Review Report: The system processes the request and displays the generated report.
7. Export Report: The user has the option to export the report in various formats (e.g., PDF, Excel).
8. Log Out: The user logs out of the reporting system.

Exceptions :-

1. Data Unavailable: If the required data is not available, the system notifies the user and suggests alternative parameters.

2. Report Generation Failure: If there is an error during report generation, the system displays an error message.

Post-conditions :-

- The report is successfully generated and available for review.
- The user has the option to export the report in the desired format.

5.22. Use Case SpecificationsMultimodal Service Booking PlatformDescription :-

- This use case describes the process by which a user books various services (e.g., transportation, accommodation) through a multimodal booking platform, allowing for seamless integration of different service types.

Actors :-

- User
- Booking System
- Service Provider

Preconditions :-

- The user has an active account and is logged into the system.
- The booking system is available and functioning properly.

Steps :-

1. Initiate Booking: The user navigates to the service booking platform.
2. Select Service Type: The user chooses the type of service they wish to book (e.g., flight, train, hotel).
3. Enter Details: The user inputs necessary details such as date, time, and location.
4. View Options: The system displays available options based on user input.
5. Select Service: The user selects a preferred service from the displayed options.
6. Confirm Booking: The user reviews the details and confirms the booking.
7. Payment: The user enters payment information and completes the transaction.
8. Receive Confirmation: The system sends a booking confirmation via email or app notification.

Exceptions :-

1. No Availability: If no services are available for the selected criteria, the system notifies the user and suggests alternatives.
2. Payment Failure: If the payment fails, the system prompts the user to retry or choose a different payment method.

3. User Authentication Failure: If the user is not logged in, the system redirects them to the login page.

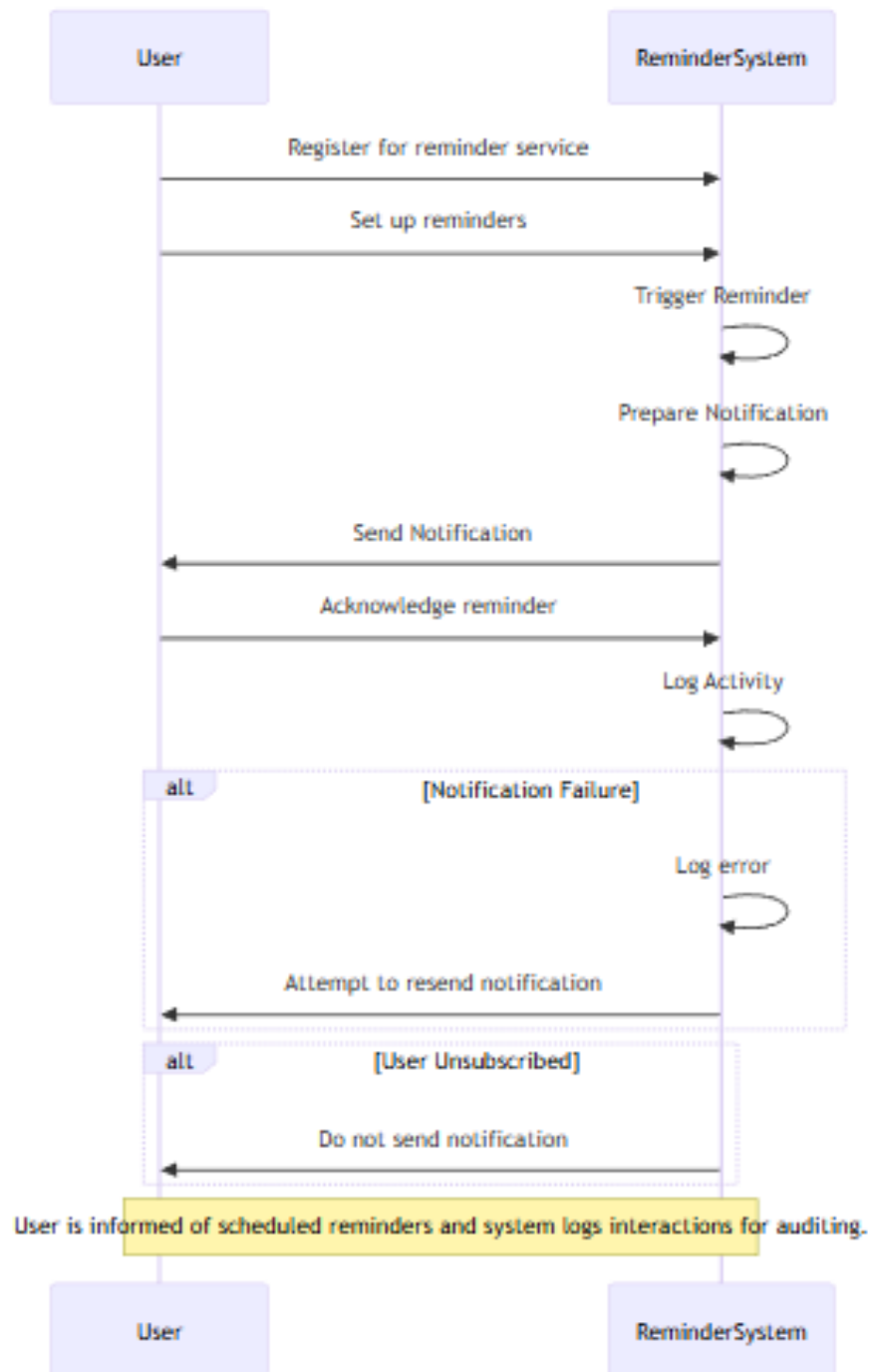
Post-conditions :-

- The service is successfully booked, and a confirmation is generated.
- The user's booking history is updated with the new service.

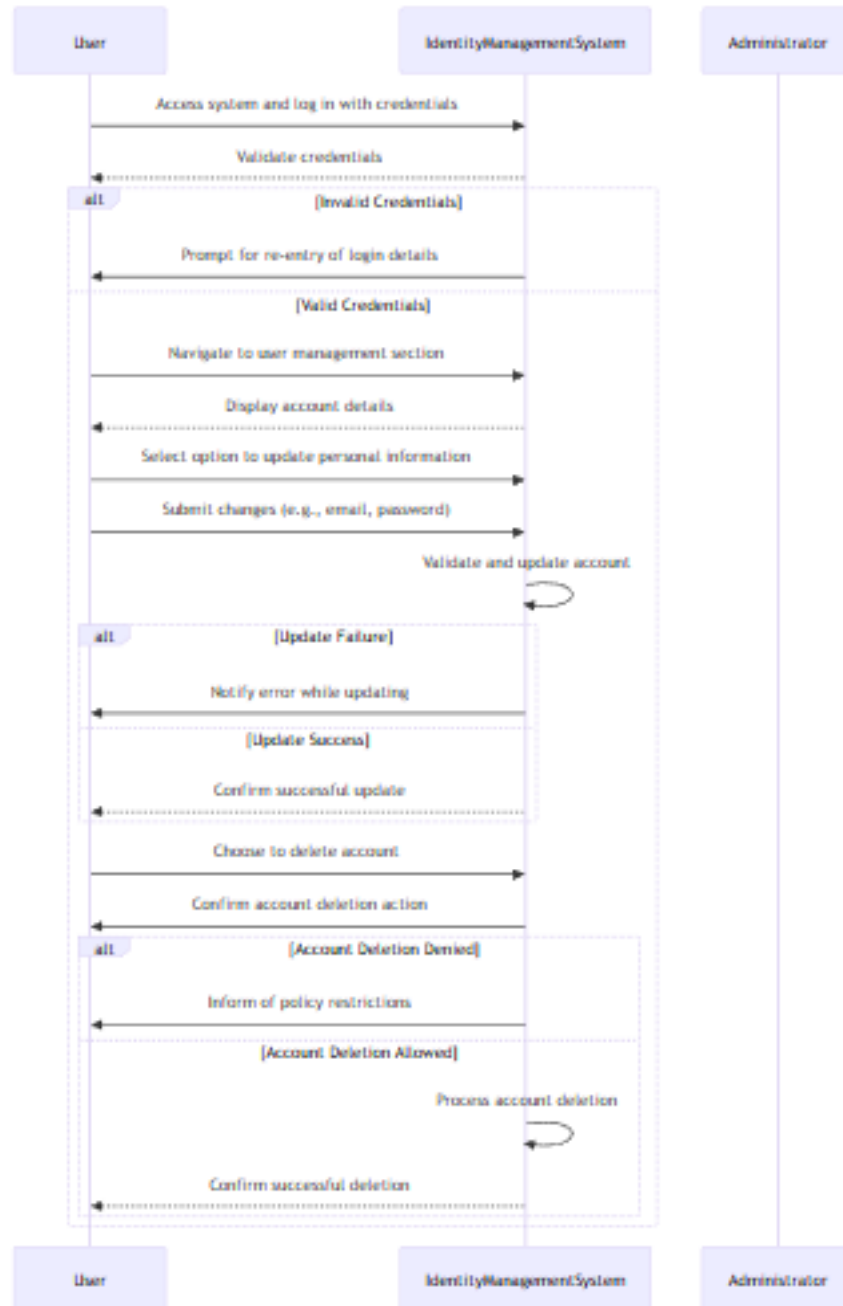
6. Interaction Diagrams

An interaction diagram visually represents the dynamic behavior and interactions between components or objects within a system during a specific use case or scenario. It captures the flow of messages, actions, and events exchanged among the system's entities to achieve a particular functionality.

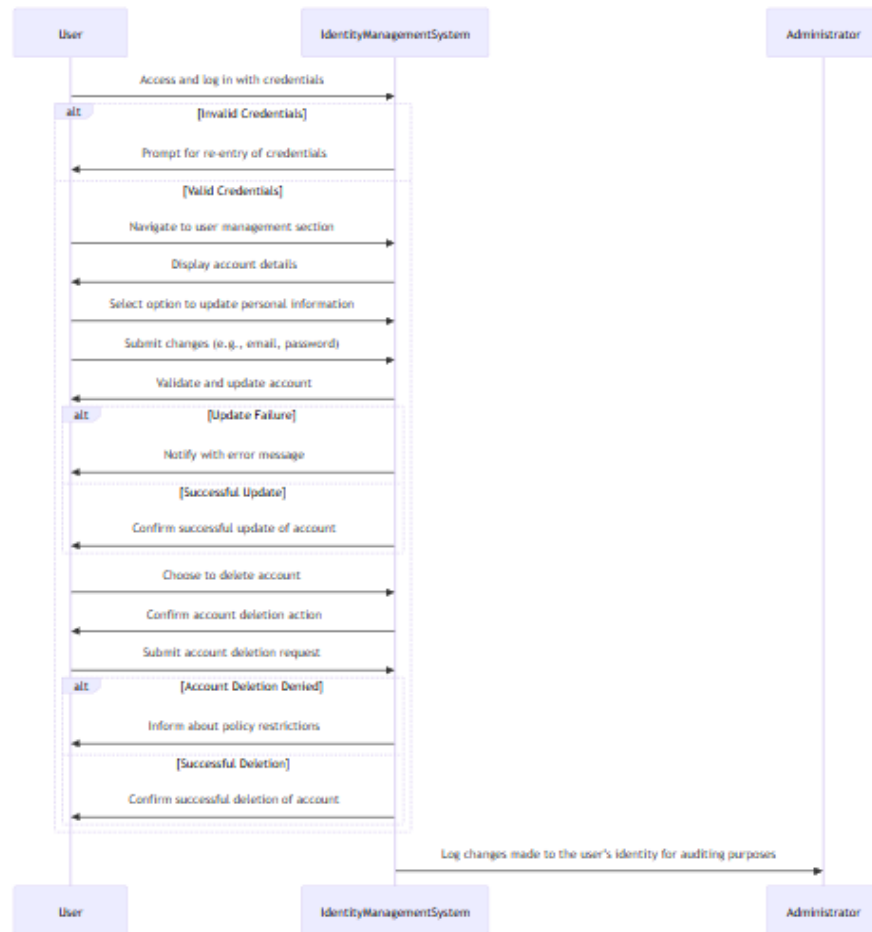
Receive Reminders



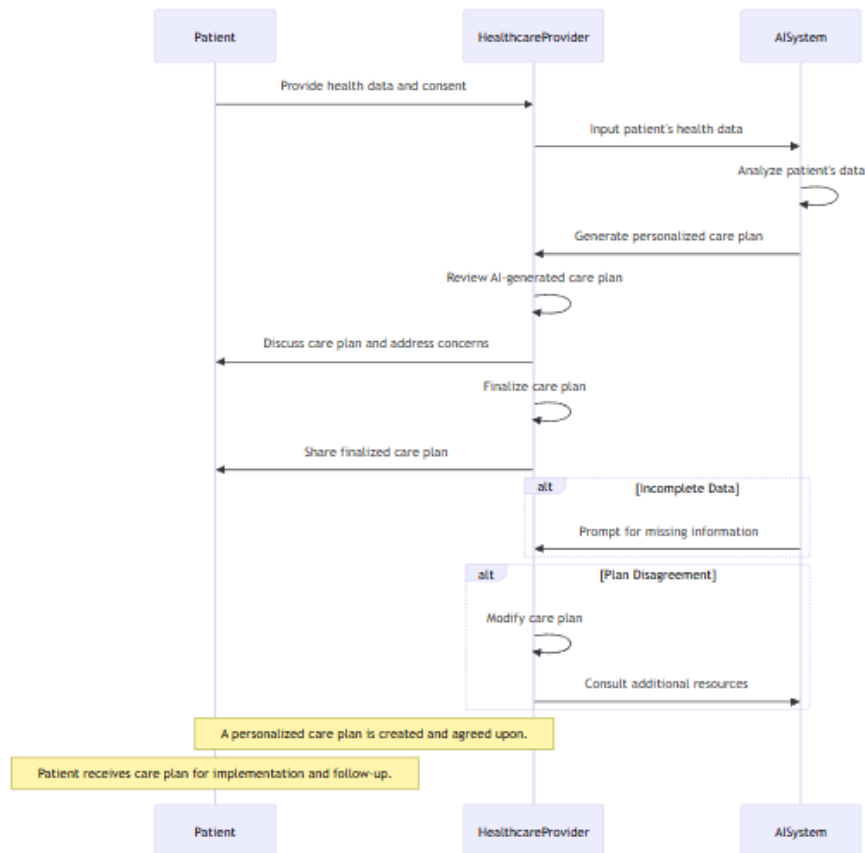
Manage User and Identity



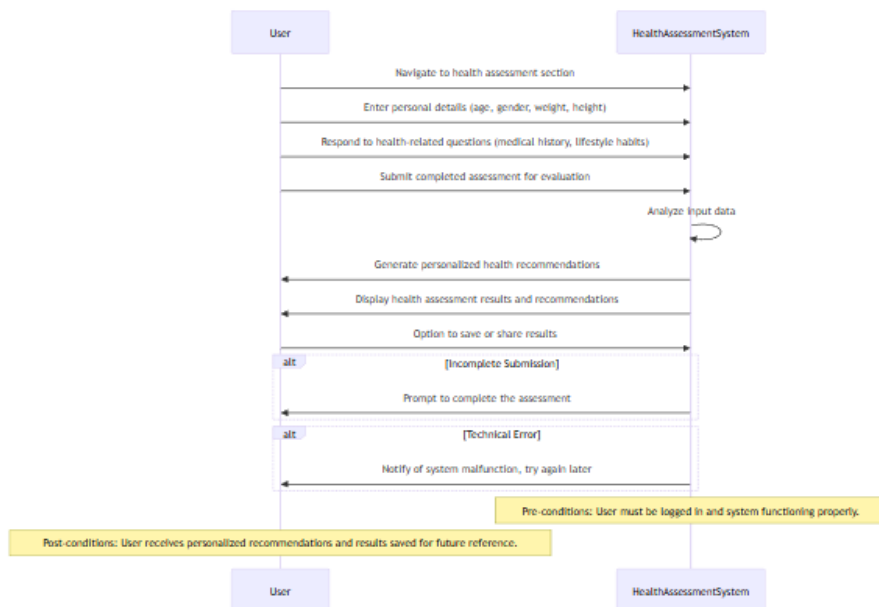
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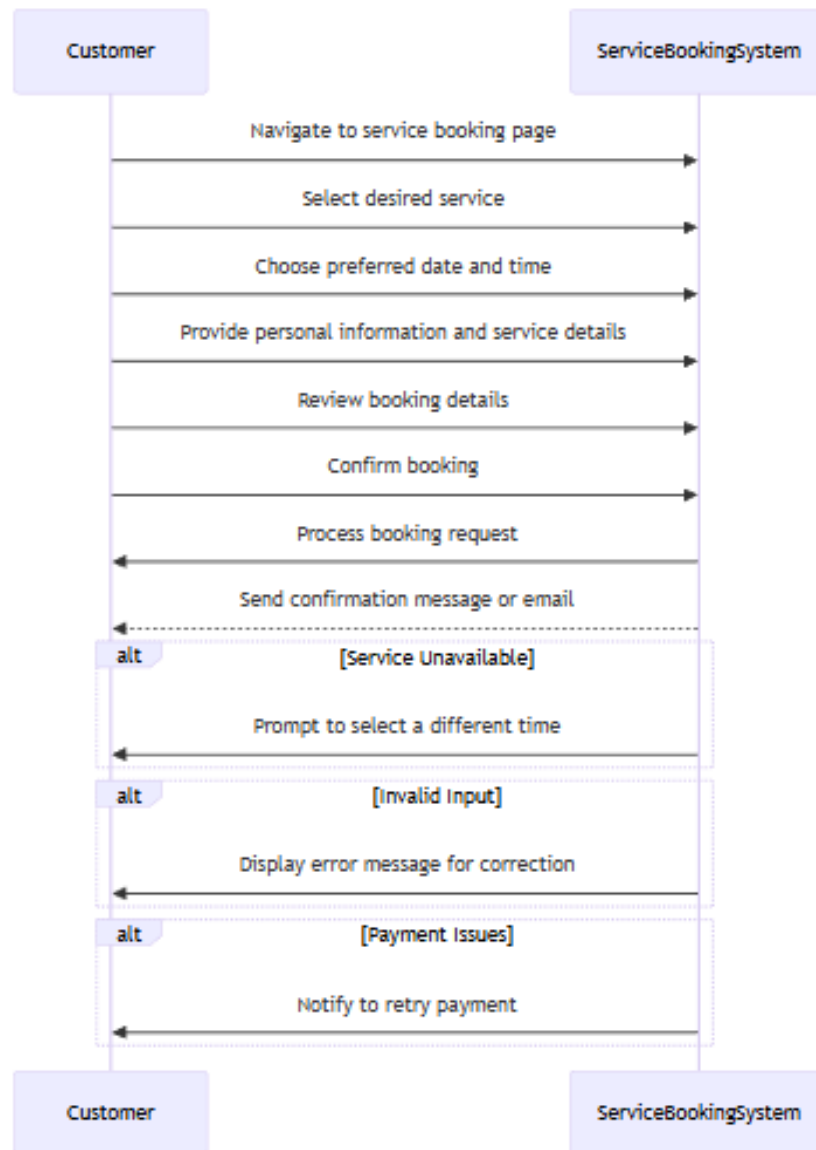
AI-driven Care Plan



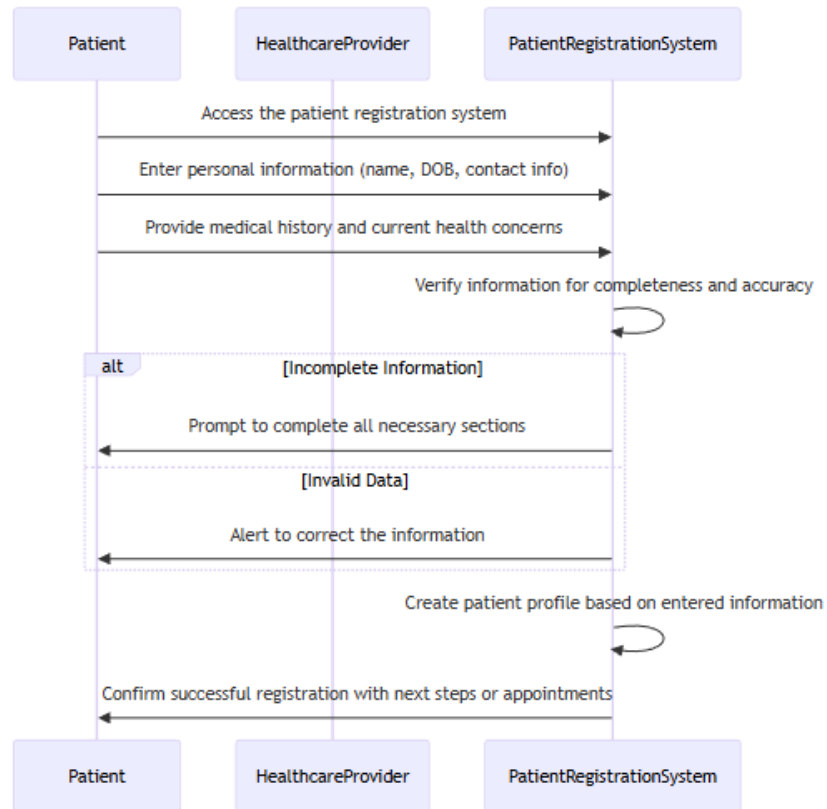
Personalized Health Assessment



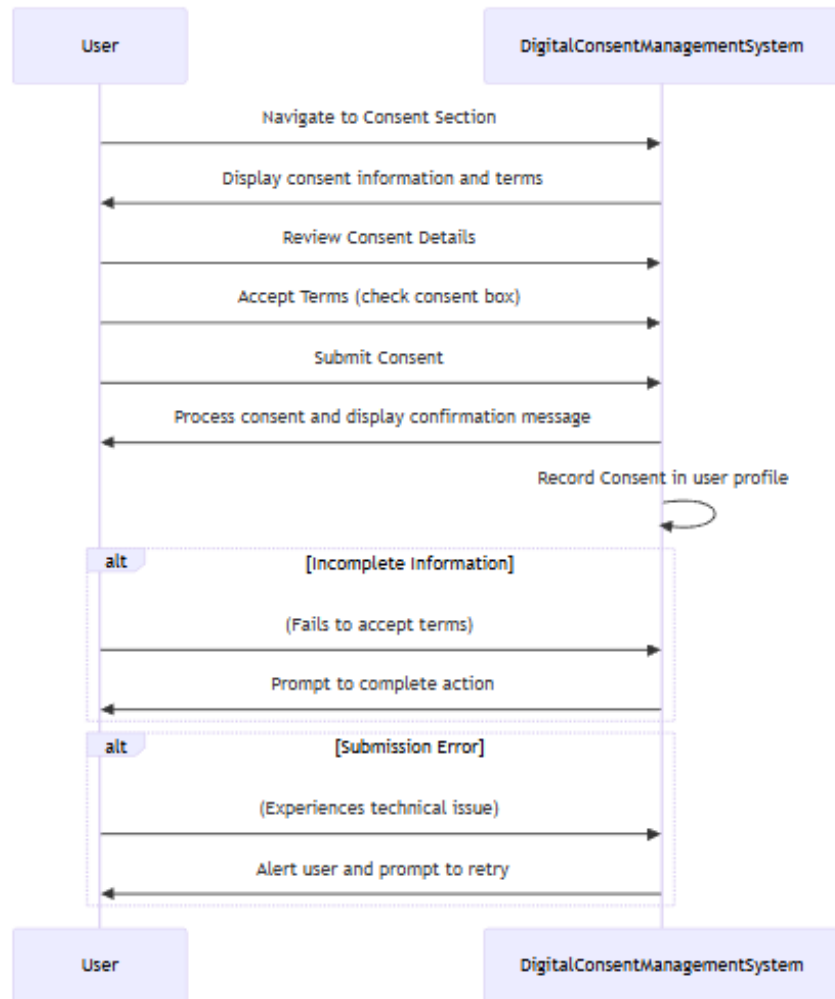
Service Booking



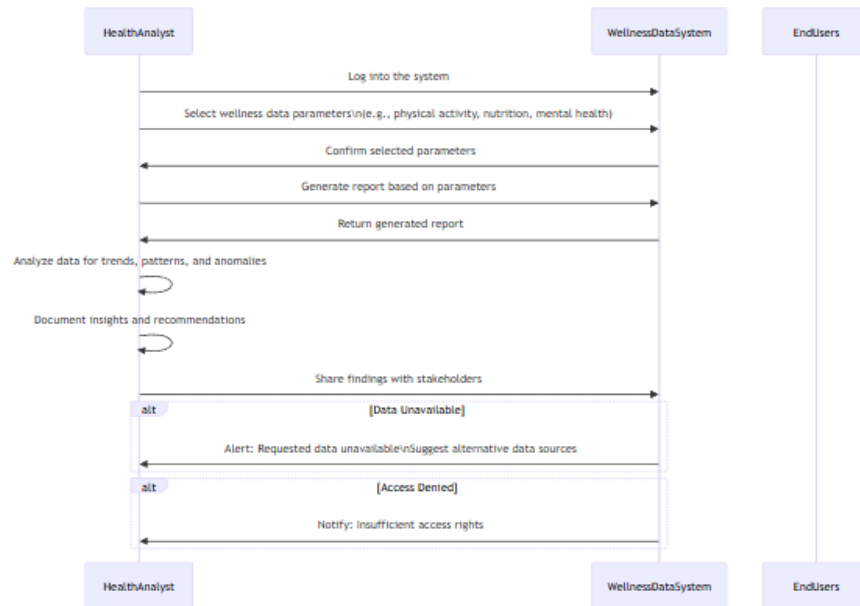
Onboard Patients



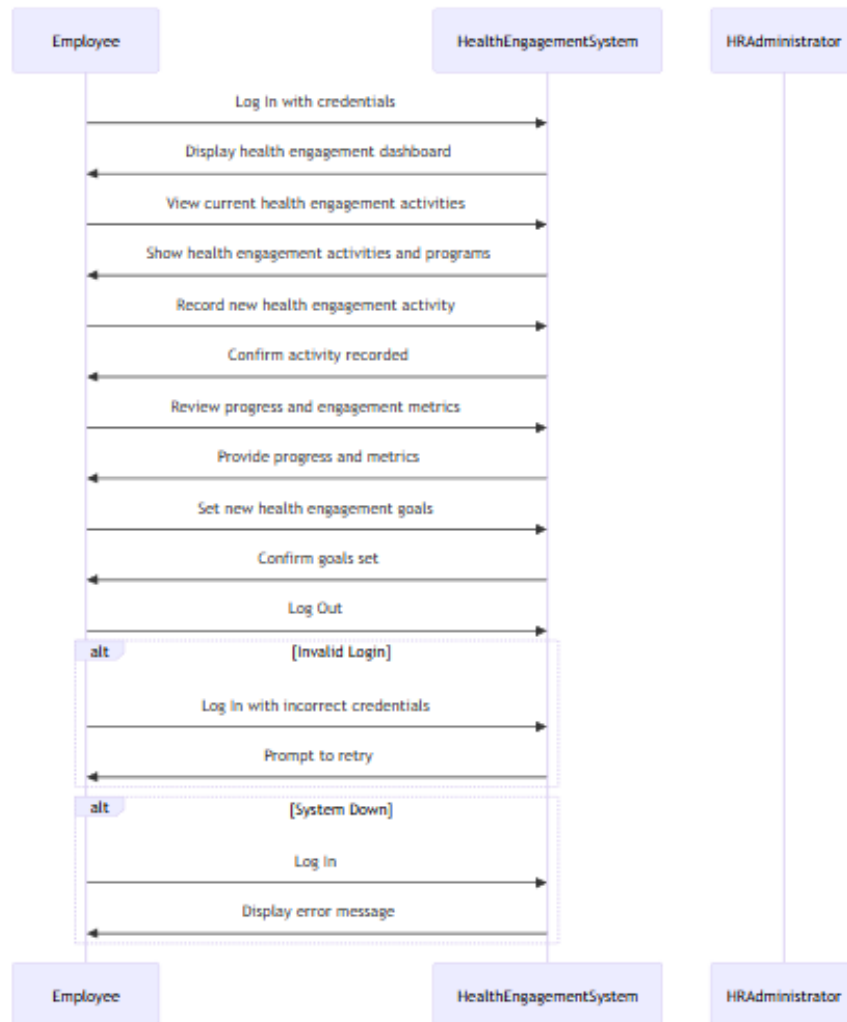
Submit Digital Consent



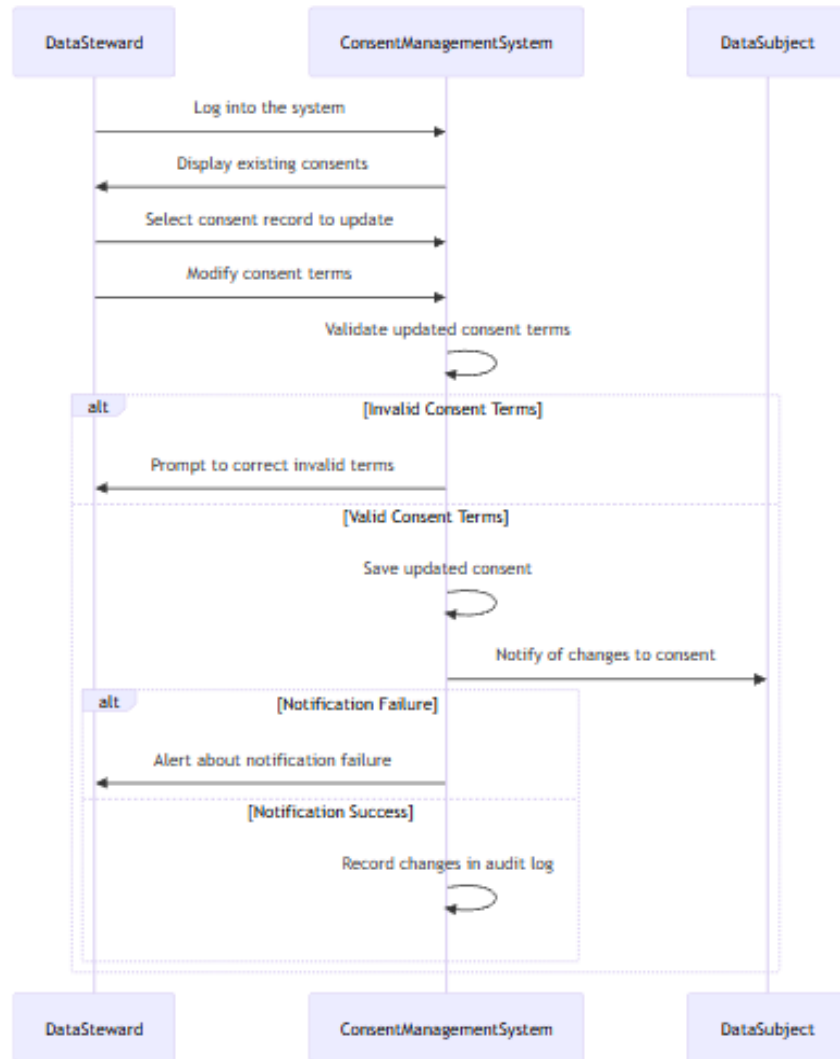
Analyze Wellness Trends



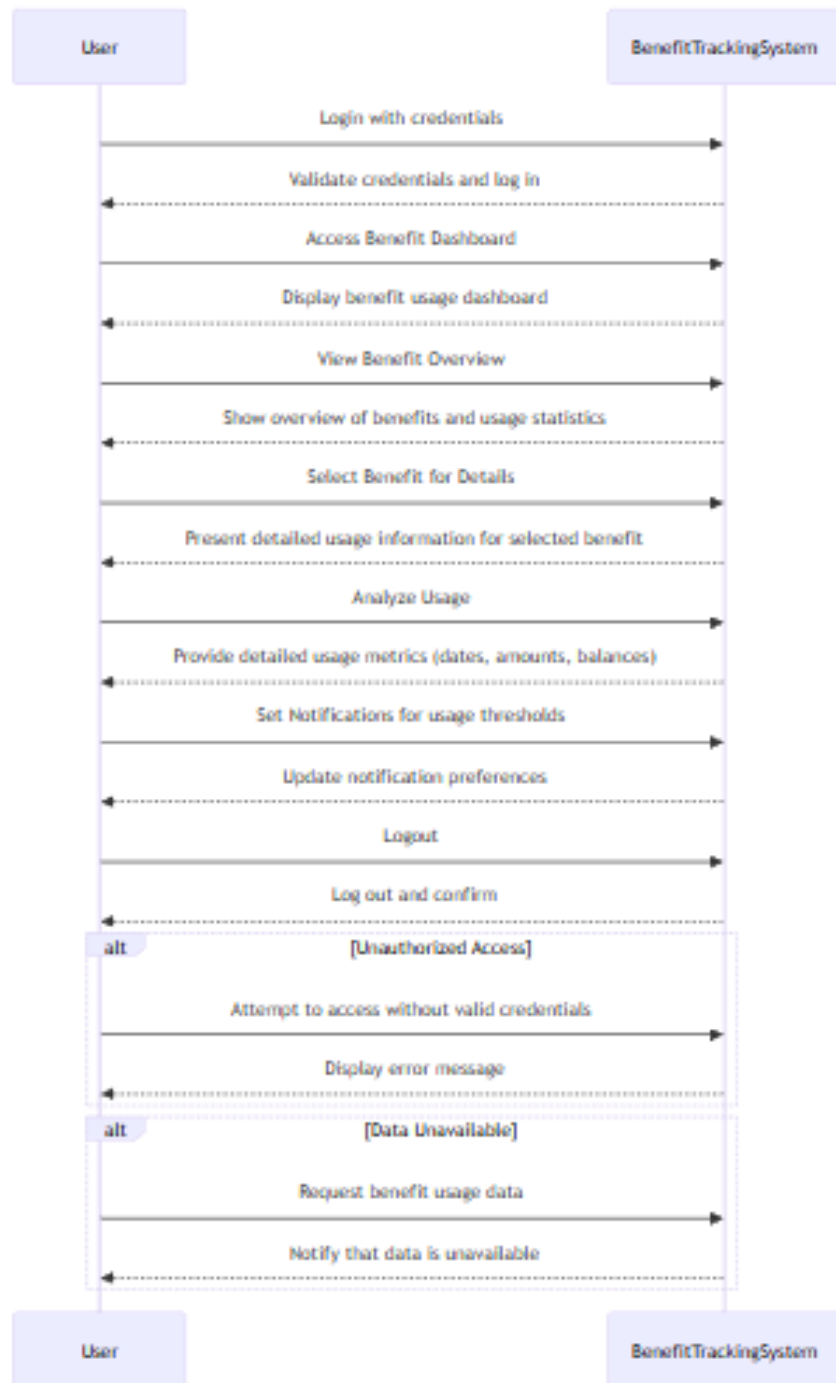
Track Employee Health Engagement



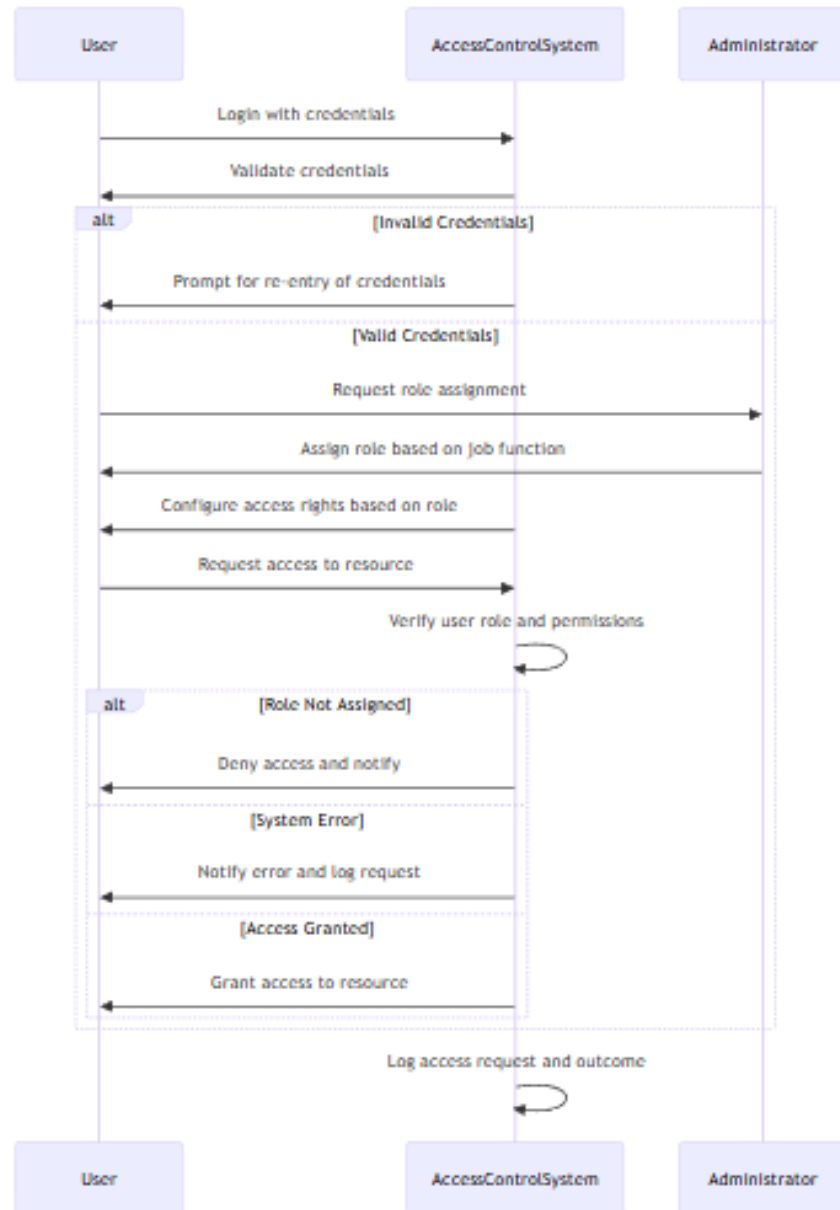
Manage Consent Governance



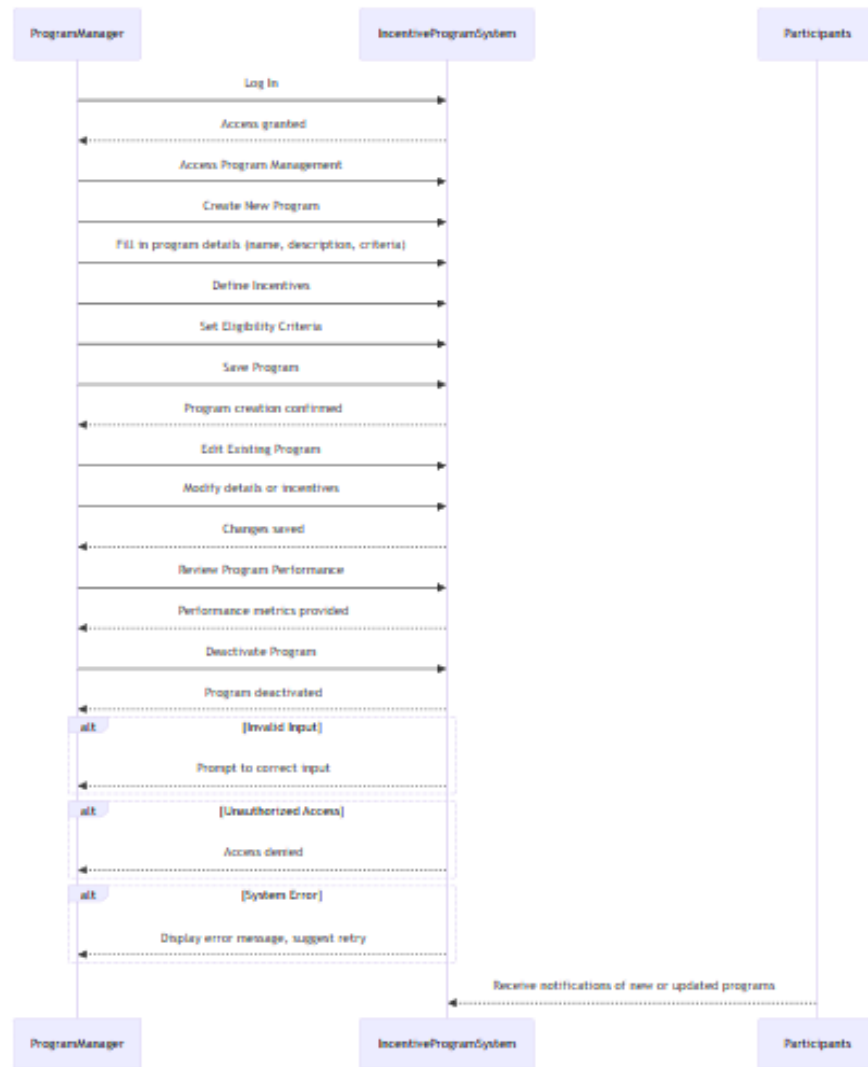
Track Benefit Usage



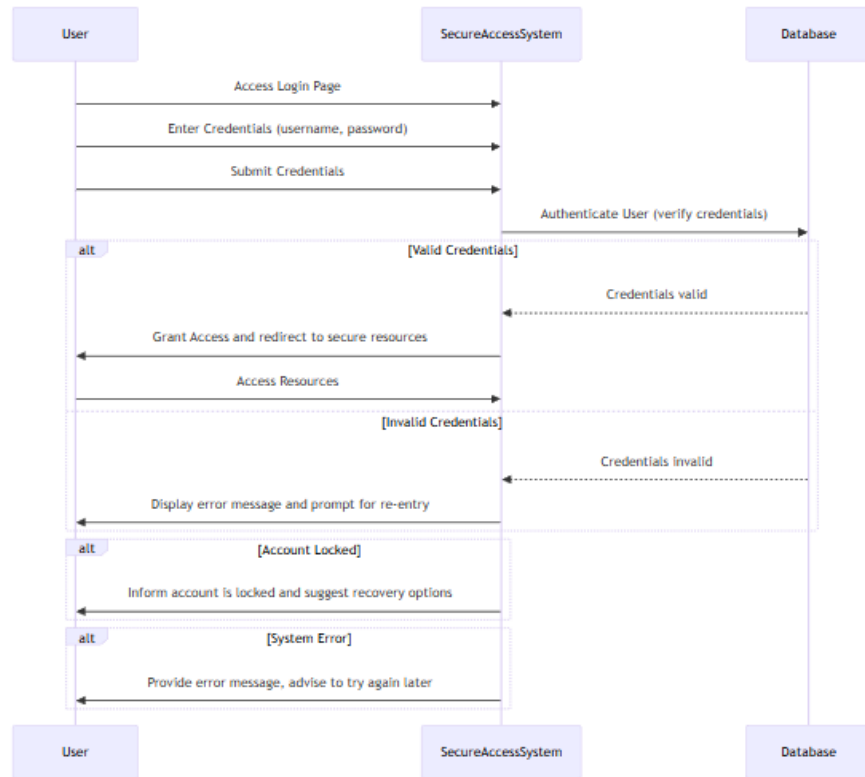
Role-Based Access Control



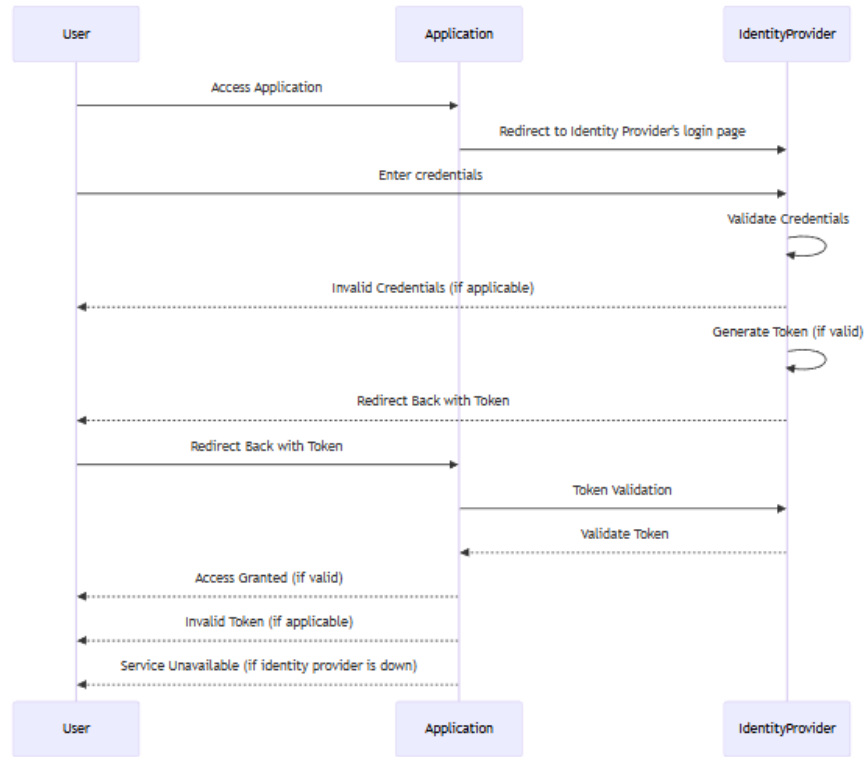
Manage Incentive Programs



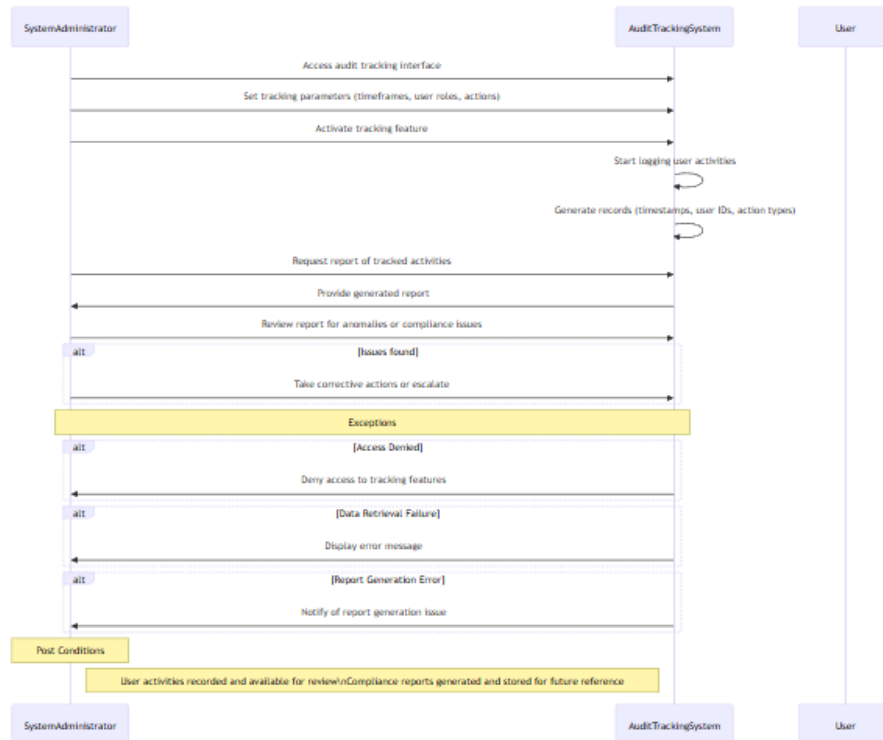
Secure Access



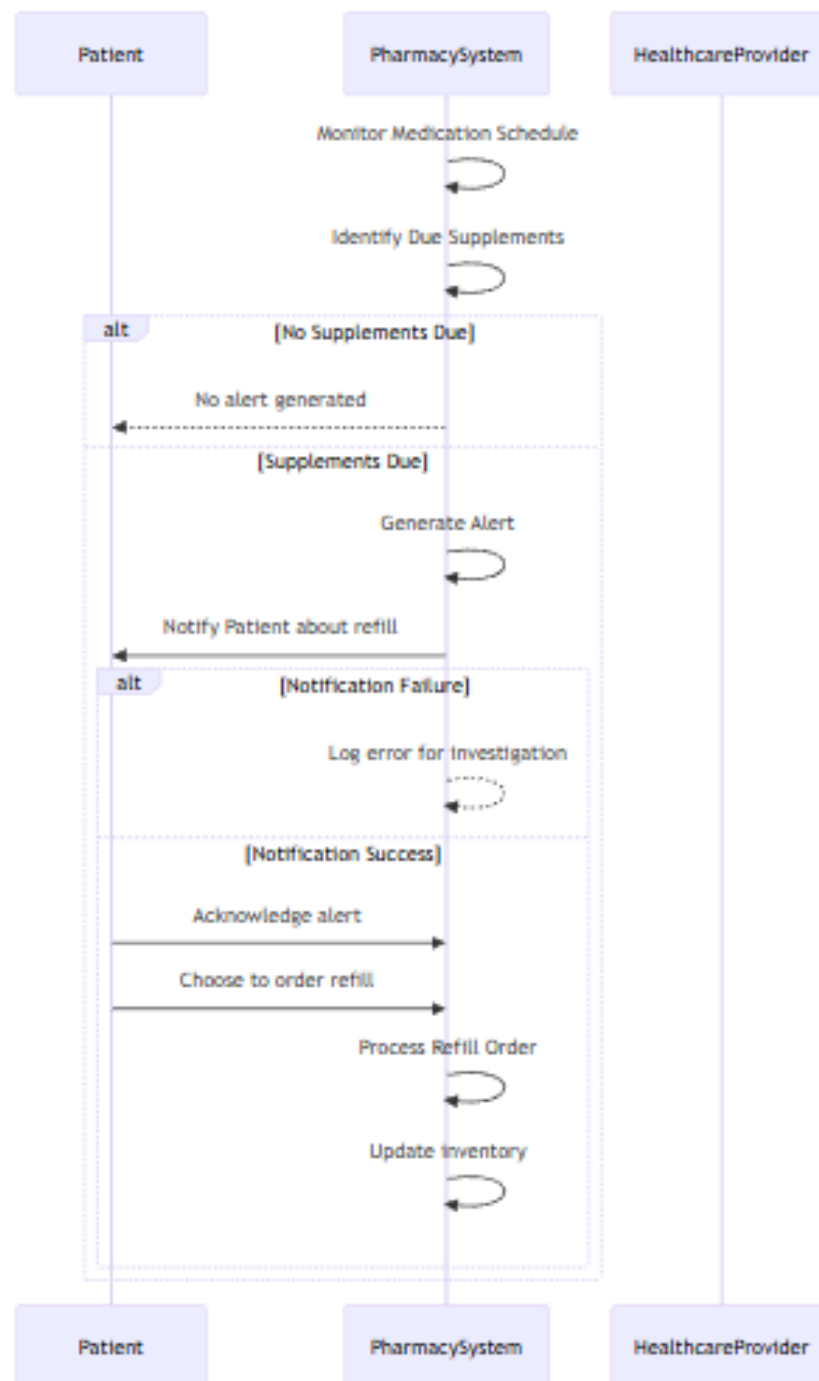
SSO Support



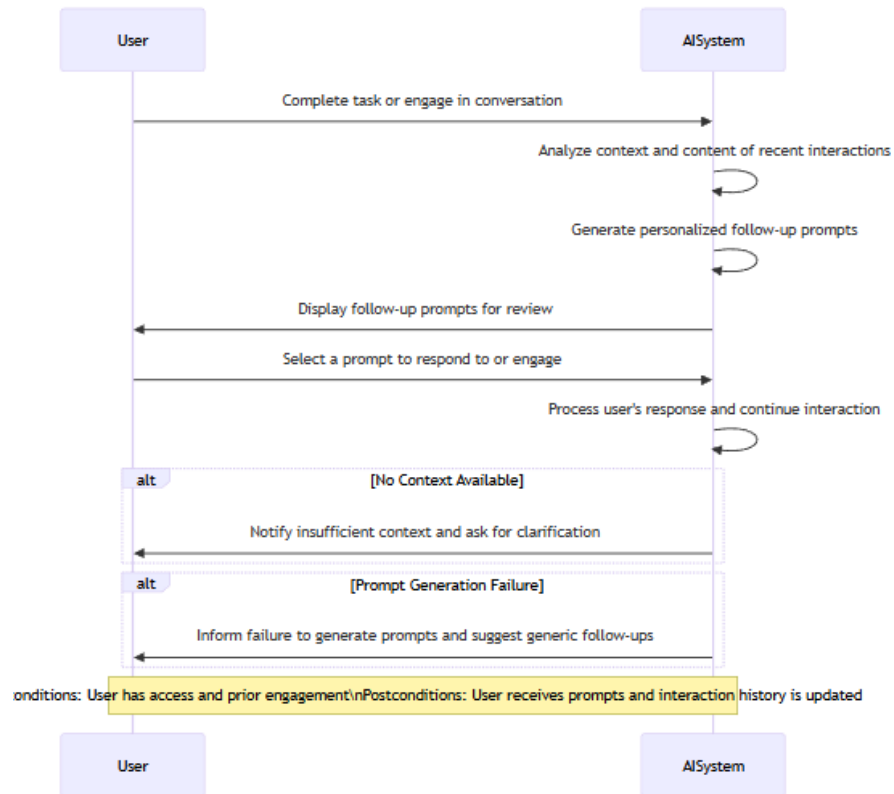
Audit Tracking



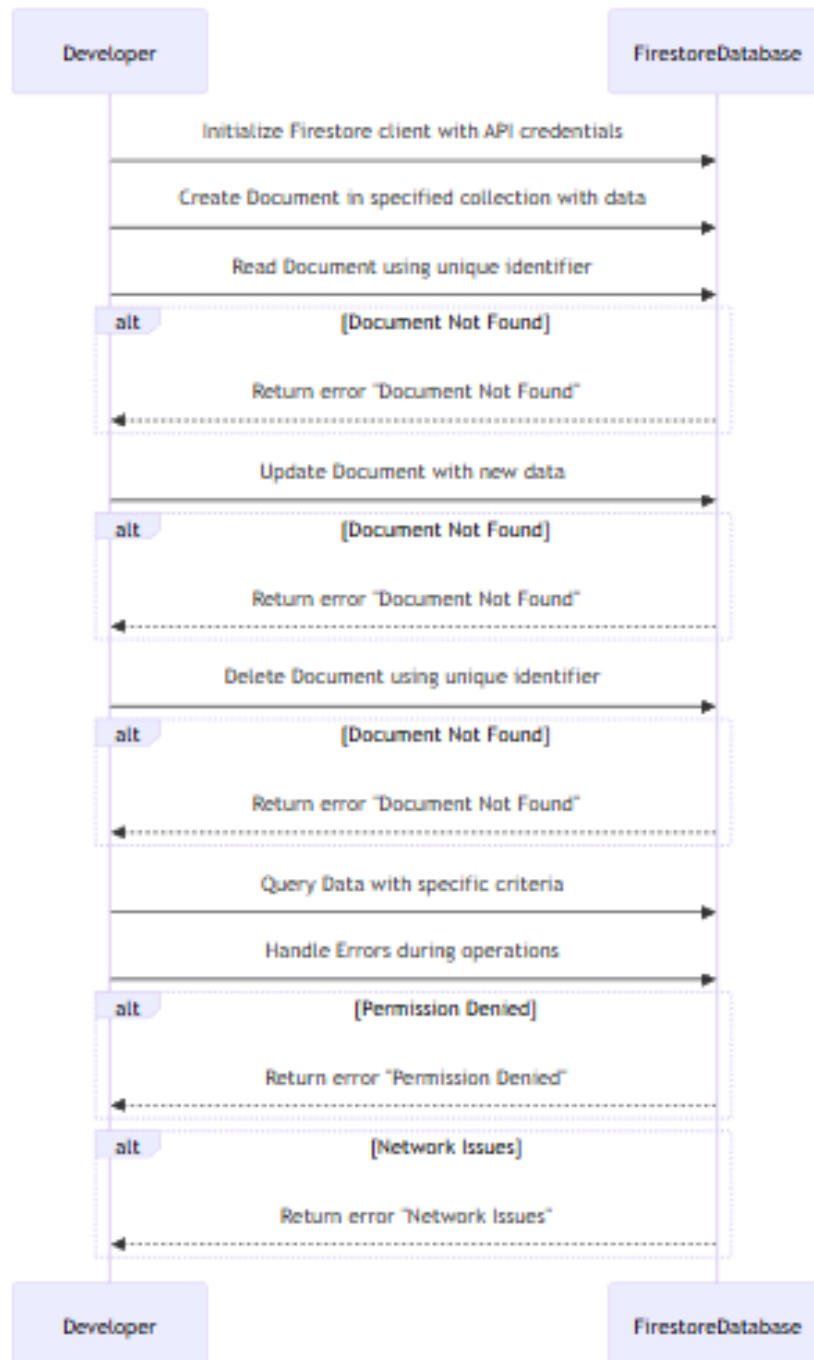
Supplement Refill Alerts



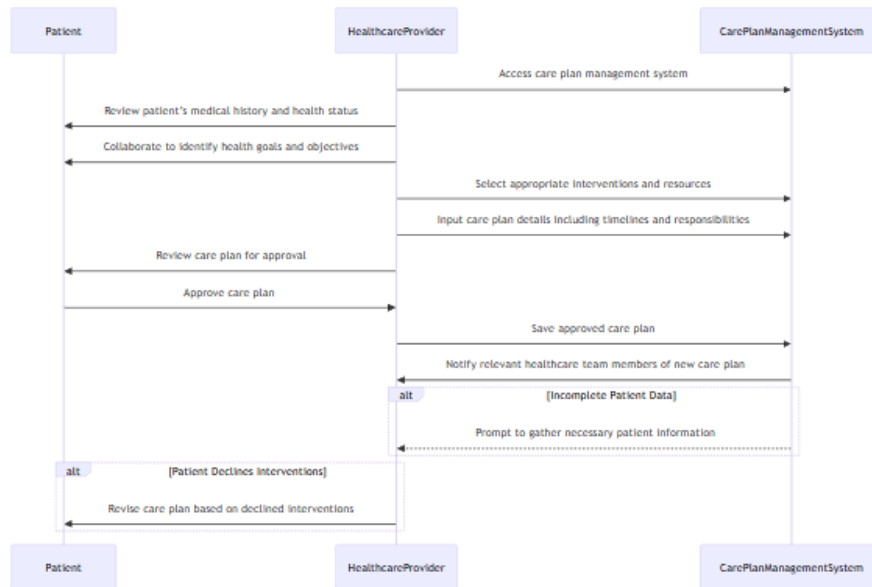
Provide AI-driven Follow-up Prompts



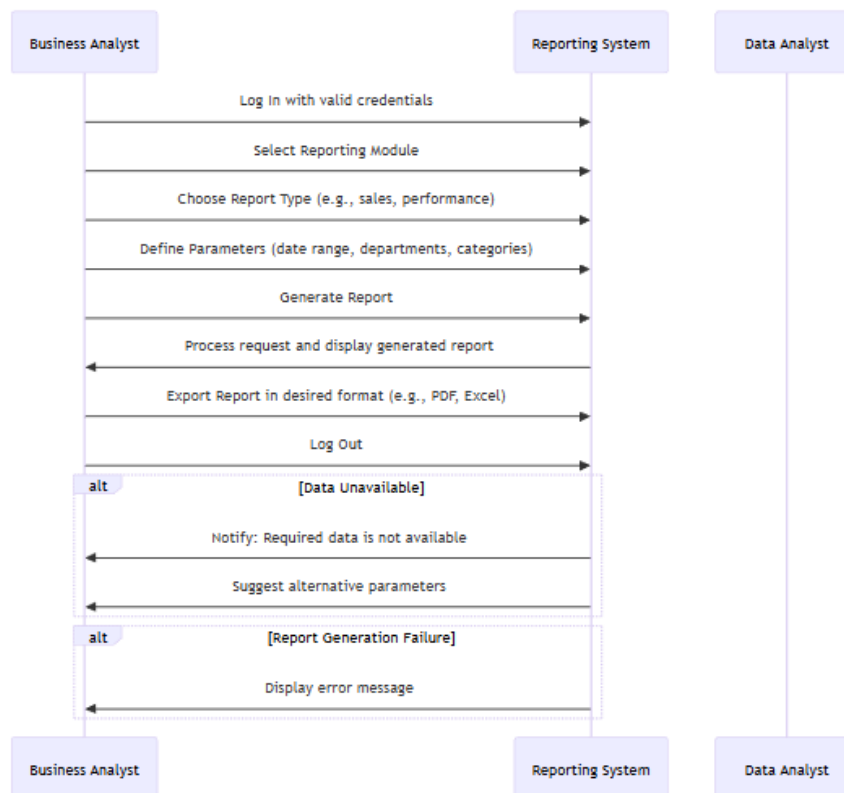
Utilize Firestore



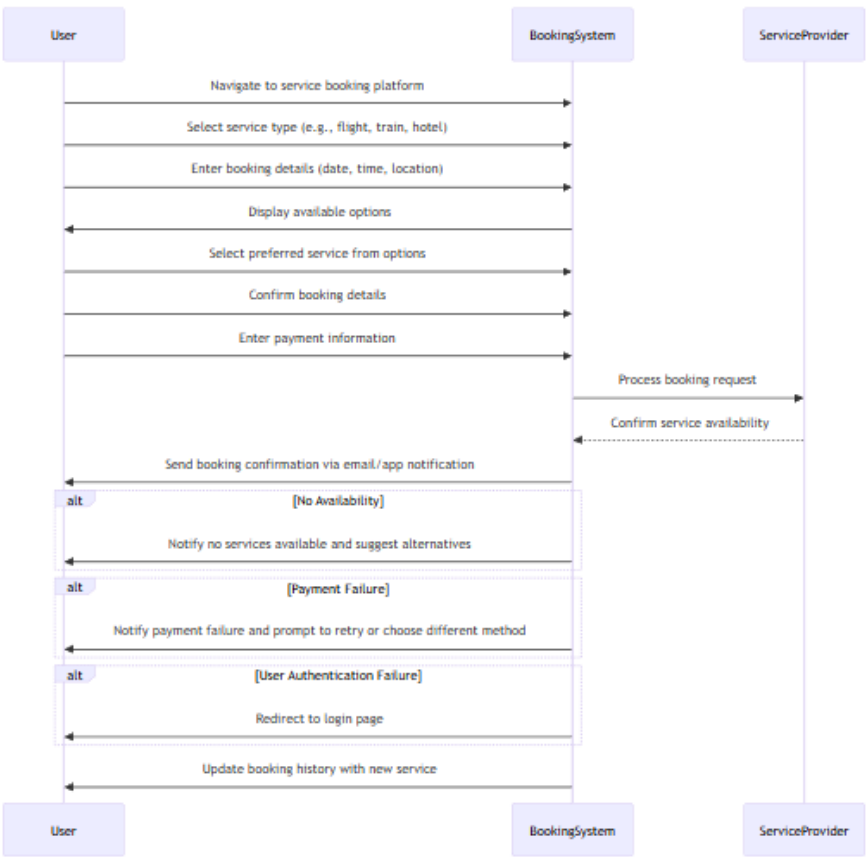
Build Personalized Care Plans



Comprehensive Reporting

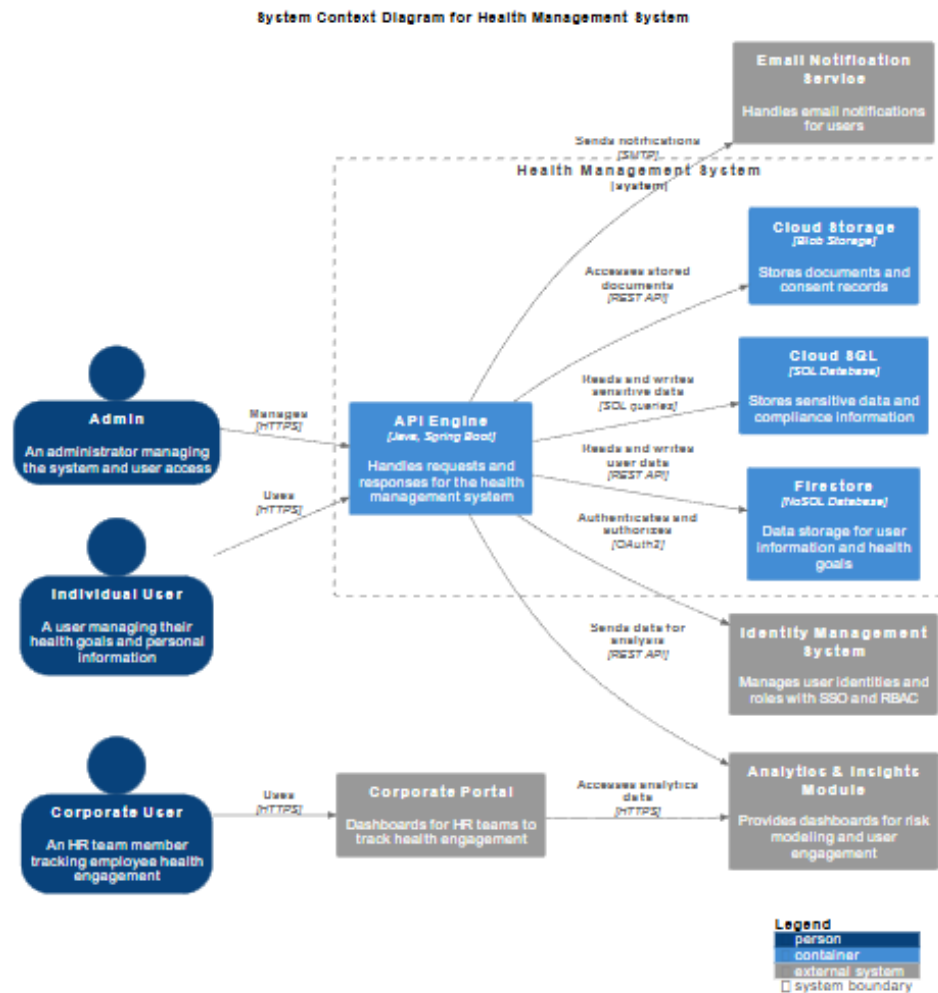


Multimodal Service Booking Platform



7. System Conceptual Diagram

A System Conceptual Diagram (SCD) is a high-level visual representation that illustrates the fundamental concepts, relationships, and boundaries of a system. It serves as a bridge between stakeholder requirements and the technical implementation details, providing a clear understanding of the system's scope and functionality without delving into specific technical solutions.



9. Data Models

9.3. Physical Data Model

A physical data model is a detailed representation of a database, showing how the data is stored, accessed, and organized within a specific database management system (DBMS). It builds on the logical data model by specifying tangible aspects like tables, columns, data types, indexes, constraints, and relationships tailored to the technical requirements and capabilities of the chosen DBMS. This model represents the actual implementation, detailing how the database will be physically constructed to optimize performance, storage, and retrieval based on factors like disk space, processing speed, and specific DBMS features. The physical data model is used by database administrators and developers as a blueprint to build and manage the actual database, ensuring that it effectively supports the application's operational requirements.



10. Deployment Models

On Prem Model

Deployment models refer to various ways in which software applications, services, or systems can be deployed or hosted, catering to different business needs and technical requirements. Common deployment models include: On-Premises, where software runs on infrastructure located within the organization's facilities, offering control and security but higher maintenance costs; Cloud-Based, where applications are hosted on cloud service providers, offering scalability and flexibility, including sub-models like public cloud, private cloud, and hybrid cloud; Public Cloud, where services are hosted on shared infrastructure provided by third-party vendors, offering cost-effectiveness but less control; Private Cloud, providing dedicated infrastructure to a single organization for more control and security; Hybrid Cloud, combining public and private cloud resources for flexibility; Multi-Cloud, using multiple cloud services from different providers to avoid vendor lock-in and enhance availability; and Community Cloud, where infrastructure is shared among several organizations with common concerns. These models help organizations choose the optimal infrastructure approach to meet their operational, financial, and strategic objectives.

